

Vantage AI — finding illegal dumps from space and pricing them in tenge

An eight-week pilot for city administrations in Kazakhstan

Team Vantage AI · 1st place, Future Minds Hackathon 2026 · hakathon-lyart.vercel.app

In short

We find illegal waste dumps in eight years of free satellite imagery, estimate the cost of each one in tenge, and turn the result into a field queue sorted by money. An inspector gets a list — where to go first, what it costs the budget, what can be recovered as recyclables — and a draft enforcement act for every site.

Sentinel and Landsat imagery is free and open. The pilot requires no servers, no procurement and no integration with internal systems.

What is already done

Northern industrial belt of Astana, a 20×20 km square, 2018–2026 archive:

Places where vegetation disappeared and did not come back	385
Left after automatic context filtering (quarries, construction, water, housing)	59
Reviewed by a person on 0.4–0.8 m imagery — all 59	15 in the field queue
— identified as dumps	7
— unclear on imagery, need a site visit	8
Waste on the identified dumps, estimate	2,500 t
Cost of removal	38.0 million ₸
Recoverable as recyclables if sorted on site	19.3 million ₸ — 51% of the removal cost
Saving of the right decision (sorting vs. plain removal)	7.6 million ₸
On the open state waste-monitoring map	1 of 15 (C00318)

No site visits have been made yet: all confirmations come from high-resolution imagery. A photo-documented visit to the fifteen sites is the first step of the pilot, and the system accepts a visit only with a photo whose GPS coordinates are next to the site.

What makes it different

- **Physics, not a black box.** Five independent signals (vegetation loss, bare soil, polymer response in SWIR, radar instability, thermal anomaly) form an evidence chain; a person makes the decision.
- **Money, not probabilities.** Monte-Carlo estimates with P10–P90 ranges; every assumption is labelled as sourced, derived or an engineering estimate.
- **Measured limits.** We know where the method fails (farmland) and can check an area before running it. Four negative results of our own models are published with numbers.

- **Complements state monitoring, does not replace it.** Every object is checked against the open state waste-monitoring map (KazEOSat-1): of our seven confirmed dumps, one is on it and six are not. The same map shows 183 dumps in our areas, and our archive search finds 18 of them — it sees where vegetation disappeared, while waste scattered over grass barely changes vegetation in a 10 m pixel and stays invisible. A second method, scanning very-high-resolution imagery in windows, finds 3–4 times more in a first test and is in development.

The pilot

Weeks	What happens	Result
1–2	Checking the territory before any processing	"the method will work here" or "it will not" — with a number
3–4	Full run over the 2018–2026 archive	candidate queue sorted by money
5–6	Review and photo-documented site visits	confirmed list and a measured removal cost
7–8	Scheduled monthly run, report	four success metrics

Success is measured in numbers: more than half of visits find a real dump; more than half of raw detections are removed without a visit; at least one site not present in any registry is found; one detection costs less than one random visit.

What we are looking for

- **City administrations** ready to run a pilot on one district.
- **Grant and accelerator support** to cover field verification and a monthly monitoring service for regional centres.
- **Data partners:** removal tariffs and registries of known sites.

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Live product: <https://hakathon-lyart.vercel.app/> · Source code: <https://github.com/k41270075-stack/hakathon>